ATUL SOMKUWAR

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EDUCATION

DEGREE	INSTITUTE	BOARD/UNIV	RESULTS	YEAR
M.Tech	Indian Institute of Technology, Bhubaneswar	IIT BBS	7.21/10	2027
B.E	Karmaveer Dadasaheb Kannamwar College of Engineering	R.T.M.N.U.	7.75/10	2021

EXPERIENCE

Map Academy of Science **Jun 21 - Aug 24**

- Delivered Mathematics and Science instruction for 2 years, strengthening students' conceptual understanding through structured and engaging teaching methodologies
- Facilitated classroom sessions using effective presentation, communication skills, and interactive teaching techniques

INTERNSHIP

Detecting crop stress before visible damage obvious **Aug 26 – Sep 26**

- Developed a deep learning model for early nutrient-stress detection in three crops across 9 classes.
- Compared MobileNetV2, EfficientNetB0, and DenseNet121 using RGB, HSV, CIELUV, and Nutrispace color spaces.
- Implemented image preprocessing, color-space conversion, and channel normalization techniques.
- Achieved **88.40% validation accuracy and 86.91% test accuracy** using CIELUV–EfficientNetB0.

PROJECTS

Fashion MNIST **Feb 26 – Mar 26**

- Engineered an image classification pipeline for 10 Fashion MNIST classes, incorporating normalization, shuffling, data augmentation, dropout, and batch size 32 for training
- Benchmarked 10 CNN architectures: Custom CNN, Modified CNN, AlexNet, LeNet, MobileNet, ResNet50, EfficientNet, ConvNeXt, DenseNet, and VGG16 to quantify and compare classification performance across diverse network architectures
- Visualized training and validation accuracy and loss using callbacks and metrics to assess convergence, evaluate model generalization and overfitting, and achieved **93.94%** test accuracy with MobileNet

ADAS EV MLP **Jan 26 – Feb 26**

- Designed a fully connected MLP pipeline for ADAS classification, incorporating data preprocessing, normalization, forward propagation, and backpropagation-based optimization
- Achieved **98.25%** test accuracy on a held-out test set across 10,000 samples spanning 15 classes, demonstrating effective generalization of the deep learning model

Predictive Modeling for Graduate Admission **Sep 25 – Oct 25**

- Evaluated Linear, Ridge, and Lasso regression models using R^2, MAE, MSE, and RMSE metrics
- Applied KNN using Euclidean-distance-based neighbor search, achieving **90% accuracy** after optimization

Voice Recognition Model **Aug 25 – Sep 25**

- Built a multiclass voice-recognition model to distinguish between 9 speakers based on voice characteristics
- Structured voice data for supervised learning by organizing speaker identities into 9 target labels, and analyzed PCA projections across PC1 and PC2 to examine speaker distribution and overlap

ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

- Won the BFKCT Season 3 Championship in the 130cc category as a member of Team Autorisors, demonstrating strong teamwork and engineering problem-solving **2019**
- Mentored M.Tech 1st-year students in Robot Design and Synthesis Laboratory (RDSL) at IIT Bhubaneswar, supporting robot design, synthesis, and hands-on laboratory experiments **2026**

Certification: ROBO AI, CREO, CATIA

Technical Skills: Python, Computer Vision, Machine Learning, Artificial Intelligence, Deep Learning, Transformers, ROS2, MATLAB